Chapter 11

Analyzing and Storing Logs

Many systems record logs of events in text files which are kept in the /var/log directory. These logs can be inspected using normal text utilities such as less and tail.

Log file	Type of Messages Stored
/var/log/messages	Most syslog messages are logged here. Exceptions include messages related to authentication and email processing, scheduled job execution, and those which are purely debugging-related.
/var/log/secure	Syslog messages related to security and authentication events.
/var/log/maillog	Syslog messages related to the mail server.
/var/log/cron	Syslog messages related to scheduled job execution.
/var/log/boot.log	Non-syslog console messages related to system startup.

- **Many programs use the syslog protocol to log events to the system. Each log message is categorized by a facility (the type of message) and a priority (the severity of the message).**
- **Available facilities are documented in the rsyslog.conf**

 # vi /etc/rsyslog.conf

```
#kern.*                                                 /dev/console

# Log anything (except mail) of level info or higher.
# Don't log private authentication messages!
*.info;mail.none;authpriv.none;cron.none                /var/log/messages

# The authpriv file has restricted access.
authpriv.*                                              /var/log/secure

# Log all the mail messages in one place.
mail.*                                                  -/var/log/maillog


# Log cron stuff
cron.*                                                  /var/log/cron

# Everybody gets emergency messages
*.emerg                                                 :omusrmsg:*

# Save news errors of level crit and higher in a special file.
uucp,news.crit                                          /var/log/spooler

# Save boot messages also to boot.log
local7.*                                                /var/log/boot.log
```

Monitoring Logs:-

```
[root@localhost ~]# systemctl stop sshd.service
[root@localhost ~]#
```

```
Sep 10 18:16:26 localhost systemd[1]: realmd.service: Deactivated succ
essfully.
Sep 10 18:16:37 localhost systemd[1772]: Starting Virtual filesystem m
etadata service...
Sep 10 18:16:37 localhost systemd[1772]: Started Virtual filesystem me
tadata service.
Sep 10 18:16:37 localhost systemd[1]: systemd-hostnamed.service: Deact
ivated successfully.
Sep 10 18:16:45 localhost PackageKit[1481]: uid 0 is trying to obtain
org.freedesktop.packagekit.system-sources-refresh auth (only_trusted:0
)
Sep 10 18:16:45 localhost PackageKit[1481]: uid 0 obtained auth for or
g.freedesktop.packagekit.system-sources-refresh
Sep 10 18:16:46 localhost systemd[1772]: Started VTE child process 247
6 launched by gnome-terminal-server process 2315.
Sep 10 18:16:51 localhost gnome-software[2009]: Only 0 apps for recent
 list, hiding
Sep 10 18:16:52 localhost gnome-software[2009]: failed to get featured
 apps: no apps to show
Sep 10 18:16:52 localhost gnome-software[2009]: Only 3 apps for curate
d list, hiding
Sep 10 18:17:04 localhost systemd[1772]: Started VTE child process 254
0 launched by gnome-terminal-server process 2315.
Sep 10 18:17:20 localhost systemd[1]: Stopping OpenSSH server daemon..
.
Sep 10 18:17:20 localhost systemd[1]: sshd.service: Deactivated succes
sfully.
Sep 10 18:17:20 localhost systemd[1]: Stopped OpenSSH server daemon.
Sep 10 18:17:20 localhost systemd[1]: Stopped target sshd-keygen.targe
t.
```

```
[root@localhost ~]# tail -f /var/log/messages
Sep 10 18:15:53 localhost systemd[1084]: Reached target Exit the Session.
Sep 10 18:15:53 localhost systemd[1]: user@42.service: Deactivated successfully.
Sep 10 18:15:53 localhost systemd[1]: Stopped User Manager for UID 42.
Sep 10 18:15:53 localhost systemd[1]: Stopping User Runtime Directory /run/user/
42...
Sep 10 18:15:53 localhost systemd[1]: run-user-42.mount: Deactivated successfull
y.
Sep 10 18:15:53 localhost systemd[1]: user-runtime-dir@42.service: Deactivated s
uccessfully.
Sep 10 18:15:53 localhost systemd[1]: Stopped User Runtime Directory /run/user/4
2.
Sep 10 18:15:53 localhost systemd[1]: Removed slice User Slice of UID 42.
Sep 10 18:15:53 localhost systemd[1]: user-42.slice: Consumed 5.462s CPU time.
Sep 10 18:15:55 localhost systemd[1]: fprintd.service: Deactivated successfully.
Sep 10 18:16:07 localhost NetworkManager[1007]: <info>  [1757508367.7428] policy
: set-hostname: set hostname to 'localhost.localdomain' (no hostname found)
Sep 10 18:16:07 localhost systemd[1]: Starting Network Manager Script Dispatcher
 Service...
Sep 10 18:16:07 localhost systemd[1]: Started Network Manager Script Dispatcher
Service.
Sep 10 18:16:10 localhost systemd[1]: systemd-localed.service: Deactivated succe
ssfully.
```

Maintaining Accurate Time

- **The timedatectl command shows an overview of the current time-related system settings, including current time, time zone, and NTP synchronization settings of the system.**

```
[root@localhost ~]# timedatectl
               Local time: Wed 2025-09-10 18:22:31 IST
           Universal time: Wed 2025-09-10 12:52:31 UTC
                 RTC time: Wed 2025-09-10 12:52:31
                Time zone: Asia/Kolkata (IST, +0530)
System clock synchronized: no
              NTP service: active
          RTC in local TZ: no
[root@localhost ~]#
```

- **A database of time zones is available and can be listed with the timedatectl list-timezones command.**

```
[root@localhost ~]# timedatectl list-timezones
Africa/Abidjan
Africa/Accra
Africa/Addis_Ababa
Africa/Algiers
Africa/Asmara
Africa/Asmera
Africa/Bamako
Africa/Bangui
Africa/Banjul
Africa/Bissau
Africa/Blantyre
Africa/Brazzaville
Africa/Bujumbura
Africa/Cairo
Africa/Casablanca
Africa/Ceuta
```

The superuser can change the system setting to update the current time zone using the timedatectl set-timezone command. The following timedatectl command updates the current time zone to Africa/Bamako.

```
[root@localhost ~]# timedatectl set-timezone Africa/Bamako
[root@localhost ~]# timedatectl
               Local time: Wed 2025-09-10 12:56:10 GMT
           Universal time: Wed 2025-09-10 12:56:10 UTC
                 RTC time: Wed 2025-09-10 12:56:10
                Time zone: Africa/Bamako (GMT, +0000)
System clock synchronized: no
              NTP service: active
          RTC in local TZ: no
[root@localhost ~]#
```

To set time manually

```
[root@localhost ~]# timedatectl set-time 11:0:0
Failed to set time: Automatic time synchronization is enabled
[root@localhost ~]# timedatectl
list-timezones    set-local-rtc    set-timezone     status
ntp-servers       set-ntp          show             timesync-status
revert            set-time         show-timesync
[root@localhost ~]# timedatectl set-ntp false
[root@localhost ~]# timedatectl set-time 11:0:0
[root@localhost ~]# timedatectl
               Local time: Wed 2025-09-10 11:00:03 GMT
           Universal time: Wed 2025-09-10 11:00:03 UTC
                 RTC time: Wed 2025-09-10 11:00:03
                Time zone: Africa/Bamako (GMT, +0000)
System clock synchronized: no
              NTP service: inactive
          RTC in local TZ: no
[root@localhost ~]#
```